

6.5.2 Maintainability

- A. The following provisions apply to all bridges with the exception of flat slab superstructures (cast-in-place or precast):
1. Design and detail superstructures using bridge bearings that are reasonably accessible for inspection and maintenance.
 2. Design bearing connections to allow for bearing replacement without having to remove the masonry plate or sole plate. Detail all steel bearing components with sliding surfaces (e.g., stainless-steel plates or PTFE plates), masonry plates and elastomeric components to be replaceable. Welded attachments are not considered a replaceable connection. For permissible steel I-girder connections, see **SDG 5.11.A**.
 3. Design and detail provisions for the replacement of bearings, such as jacking locations, jacking sequence, jack load, etc. Verify that the substructure and superstructure widths are sized and detailed to accommodate the jacks and any other required provisions. Simple span pretensioned beams are exempt from this requirement.
- B. Separate details and notes describing jacking procedures are required for steel girder and segmental concrete box girder bridges. For steel I-girder bridges, design, and detail so that jacks can be placed directly under girder lines. For steel and segmental concrete box girder bridges, design, and detail so that jacks can be placed directly under pier diaphragms. Always include a plan note stating that the jacking equipment is not part of the bridge contract.

Commentary: Few concrete I-beam bridges have required elastomeric bearing pad replacement. Occasional replacement of these pads does not justify requiring these provisions for every bridge.

6.5.3 Lateral Restraint

- A. Determine if lateral restraint of the superstructure of a bridge is required and make necessary provisions to assure that the bridge will function as intended. These provisions include considerations for the effects of geometry, creep, shrinkage, temperature, and/or seismic on the structure. When lateral restraint of the superstructure is required, develop the appropriate method of restraint as described hereinafter.
- B. Provide lateral restraint as follows for superstructures supported on elastomeric bearings:
1. For concrete beam or girder superstructures, when the required restraint exceeds the capacity of the bearing pad under the **LRFD** Table 3.4.1-1 Extreme Event I or II load combinations, provide concrete blocks (shear lugs) cast on the substructure and positioned to not interfere with bearing pad replacement. Do not use concrete blocks (shear lugs) to resist lateral forces from other load combinations.